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SATURDAY, MAY 29, 1886.

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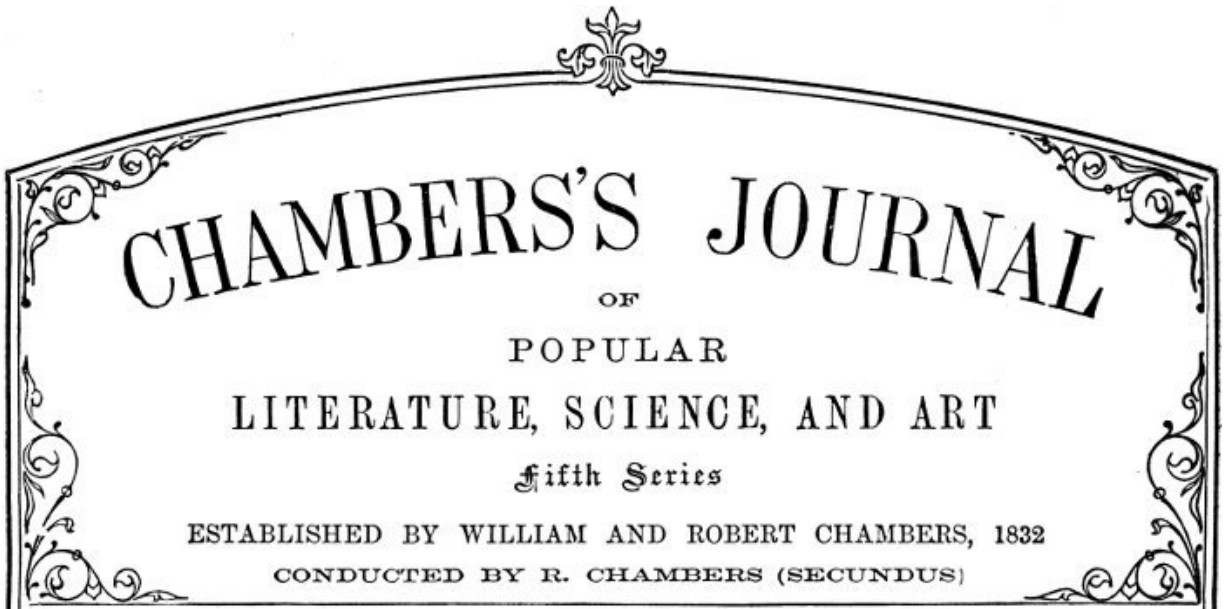
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Haflidi Steinsson, the priest here mentioned, had long since gone back to Iceland, where he died parish priest of Breidabolstad; and in chronicling his death, the annalist adds that ‘he was King Eirik’s chaplain at the time that his daughter Margaret was taken to Scotland, as she herself afterwards bore witness when she was being carried to execution at Nordness.’ Indeed, so prevalent was the belief in the personal identity of the Claimant with the daughter of King Eirik who died on the voyage to Orkney in 1290, that the place of her execution became a resort of pilgrims; and many of the priesthood having countenanced the popular belief in her martyrdom, a chapel was built on the spot where she suffered; and though uncanonised, and reprobated by the dignitaries of the church, her memory was held in reverence till the Reformation as St Maritte (Margaret), the Martyr of Nordness. In 1320, the number of pilgrimages to this irregular shrine had become so numerous that Bishop Audfinn of Bergen issued an official interdict against them, an interference which was resented by his canons, some of whom were bold enough to protest against its promulgation.

Nothing is known of the Claimant’s previous history, except that the contemporary annalist states that she came to Bergen in a ship from Lübeck. Absolom Pedersen asserts that she came from Scotland, but gives no authority for the statement, and there is sufficient evidence in the records to render this highly improbable. But it is a very remarkable circumstance that Wyntoun, the popular historian of his time, gave currency in Scotland to the statement—which we must assume to have been then the popular belief—that the Maid of Norway was put to death in her own country by martyrdom. After giving circumstantial details of the sending of the Scottish embassy to Norway, consisting of Sir David of the Wemyss and Michael Scot of Balwearie, he adds, that when they arrived—

Dead then was that Maiden fair,
That of law suld have been heir,
And appearëd til have been
By the law of Norway Queen;
But that Maiden sweet for-thi [*therefore*]
Was put to death by martyr.

In accordance with the usage of the period, the expression of the chronicler describing the manner of her death would be universally understood to mean burning at the stake; and the evident anachronism, as well as the inherent improbability of the narrative, is accounted for by the fact that it quite accurately describes the death of the Claimant, but assigns it to the time of the death of the Princess. The reason given by Wyntoun for the 'martyrdom' is, that the Norwegians—though their law allowed—could not brook the idea of a woman succeeding to the crown; and this also may be accounted for by the fact that the woman who suffered was a pretender to the crown.

No incident in Scottish history is more pathetic than that of the untimely death of the young Princess on her voyage to Orkney; and no single event in the whole course of that history has exercised a more important influence on the destinies of the nation. In these circumstances, we cannot cease to wonder how it came to pass that there is no authentic record of its details in the contemporary or nearly contemporary chronicles of Scottish or Norwegian history. The only contemporary document in Scottish record which notices her death is the letter of the Bishop of St Andrews to Edward I., dated at Leuchars the 7th of October 1290, in which the bishop states that there was a rumour of her death; but that he had heard subsequently that she 'had recovered of her sickness, but was still weak.' It was plain, however, that the bishop did not believe the rumour of her recovery, for he concludes his letter by praying King Edward to approach the Borders with his army to prevent bloodshed, seeing that Sir Robert Bruce had come to Perth and Mar and Athole were collecting their forces. On the Norwegian side, there is a total absence of authentic contemporary record of the time and manner of the death of the Princess; and there would have been absolutely nothing known of the details of her decease, if it had not been for the appearance ten years later of the Claimant, whom Munch, the historian of Norway, following Bishop Audfinn, has no hesitation in designating 'The False Margaret.'

In his official interdict of 1320, forbidding the people 'any longer to invoke that woman with great vows and worship as if she had been one of God's martyrs,' the bishop states that he has deemed it his duty to declare the truth as to her case; 'She said, indeed, that she was the child and lawful heir of King Eirik; but when she came from Lübeck to Bergen she was gray-haired

and white in the head, and was proved to be twenty years older than the time of King Eirik's marriage with Margaret of Scotland. He was then only thirteen winters old, and consequently, could not have been the father of a person of her years. And then he had no other child than one daughter by Queen Margaret. This only child of King Eirik and Queen Margaret was on her journey to Scotland, when she died in Orkney between the hands of Bishop Narve of Bergen, and in the presence of the best men of the land, who had attended her from Norway; and the bishop and Herr Thore Hakonson and others brought back her corpse to Bergen, where her father had the coffin opened and narrowly examined the body, and himself acknowledged that it was his daughter's corpse, and buried her beside the queen her mother, in the stone wall on the north side of the choir of the cathedral church of Bergen.'

Although we owe these details of the Princess's death and burial, meagre as they are, to the bishop's anxiety to confute the pretensions of the Claimant, there can be no room for doubt as to their strict truth. And yet it was possible, ten years after the event, for a Claimant so to influence the popular belief, that, although burned to death as a traitorous impostor, she was regarded by many of the priesthood as a martyr; and by the common people was not only worshipped as a saint in the church erected to her memory on the spot where she suffered, but celebrated in songs which long continued to be handed down among them. Even to this day, the precise date of the death of the Princess Margaret remains unknown; and until quite recently, it was generally believed that she had been buried in Kirkwall Cathedral, as is indeed stated by the Danish archæologist Worsaae in his account of that edifice given in his work on *The Danes and Northmen in England*. No *History of Scotland*, until the issue of the last edition of Dr John Hill Burton's, has noticed the curious episode of the False Margaret, a knowledge of which is necessary in order to account for the fact that, in Wyntoun's time, it was the popular belief in Scotland that the Maid of Norway had suffered martyrdom at the hands of her own countrymen.

It is curious that in connection with the history of Scotland, and before the close of the fourteenth century, we find the story of another Claimant not less audacious in his pretensions, but much more fortunate in his patrons, by whom he was maintained till his death as a state pensioner, and buried in one of the churches of Stirling under the royal name and regal title to which

he had laid claim. There was this strange element in his case that he was the second personator of the same dead king. Readers of English history are familiar with the incidents of the revolution which placed Henry of Lancaster on the throne, and consigned 'the good King Richard' to a perpetual prison in Pontefract Castle. But the subsequent events in the life of the imprisoned monarch, and the date and manner of his death, are shrouded in an impenetrable obscurity. One of the ablest of our Scottish historians, Patrick Fraser Tytler, has even declared, after an elaborate investigation of the whole available evidence, that this second Claimant, whose story we are about to notice, was Richard II. in reality.

It is well known that shortly after the king's imprisonment, there was a conspiracy to replace him on the throne. The conspirators attempted to attract the people to their cause by spreading abroad the rumour of his escape from Pontefract; and, as is stated by a contemporary chronicler, 'to make this the more credible, they brought into the field with them a chaplain called Father Maudelain, who so exactly resembled good King Richard in face and person, in form and speech, that every one who saw him declared that he was their former king.' The conspiracy failed; and those most deeply concerned in it, among whom was the first personator, Father Maudelain, were beheaded.

Shortly afterwards, it was given out that King Richard had died in Pontefract Castle, on or about the 14th of February 1399. Rumour, indeed, spoke freely of the suspicion, that if he were dead, he had surely been murdered by his enemies, and with the connivance of the reigning king. It was not till nearly a month after the alleged date of his death that, in order to silence the popular rumours, King Henry caused the body to be brought publicly to London 'with the face exposed,' and laid in state for two days in the church of St Paul, 'that the people might believe for certain that he was dead.' 'But,' says the old chronicler formerly quoted, 'I certainly do not believe that it was the king, but I think it was Maudelain, his chaplain,' who had been beheaded little more than a month previously.

There were many who shared this unbelief; and in 1402, the rumours that King Richard was yet alive became so persistent and circumstantial, that King Henry dealt with them by putting to death a number of persons, principally priests and friars, for spreading such treasonable reports. The cause of the revival of these rumours at this time is revealed in a document

issued by King Henry, requiring the sheriffs to arrest all persons guilty of spreading the report that King Richard was alive, which had arisen from a person calling himself King Richard having appeared in Scotland in company with one William Serle, who had been groom of the robes to Richard, and had possessed himself of his signet.

As the scene thus shifts to Scotland, we naturally turn to the Scottish chronicles and records for the further elucidation of the mystery. Wyntoun and Bower—each writing of events which happened within his own lifetime—narrate the story of this second Claimant in much the same manner. He came from the out-isles of Scotland, having been discovered in the kitchen of Donald, Lord of the Isles, by persons who had seen King Richard, and recognised his likeness. He was sent in charge of Lord Montgomery to Robert III. of Scotland, by whom he was well received, and assigned a pension of one hundred merks yearly. After King Robert's death, the pension was continued by the Regent Albany. The Scottish Chamberlain, in charging his accounts with these annual payments, has entered them as paid to 'King Richard of England.' Finally, we learn from an old Scottish chronicle that when he died at Stirling in 1419, his body was buried on the north side of the high-altar of the Church of the Preaching Friars, and a long Latin epitaph graven over his tomb informed the reader that 'Here lies buried King Richard of England.' Yet it has been established as clearly as any such question can now be established by evidence, that this second personator of King Richard was an adventurer named Thomas Ward, or Thomas of Trumpington, who, with his confederate William Serle, is exempted by name from the general amnesty granted to political offenders by Henry IV. in 1403.

IN ALL SHADES.

CHAPTER XXVIII.

MEANWHILE, Harry Noel himself was quite unconsciously riding round to the Hawthorns' cottage, to perform the whole social duty of man by Edward and Marian.

'So you've come out to look after your father's estates in Barbadoes, have you, Mr Noel?' Marian inquired with a quiet smile, after the first greetings and talk about the voyage were well over.

Harry laughed. 'Well, Mrs Hawthorn,' he said confidentially, 'my father's estates there seem to have looked after themselves pretty comfortably for the last twenty years, or at least been looked after vicariously by a rascally local Scotch agent; and I've no doubt they'd have continued to look after themselves for the next twenty years without my intervention, if nothing particular had occurred otherwise to bring me out here.'

'But something particular did occur—eh, Mr Noel?'

'No, nothing occurred,' Harry Noel answered, with a distinct stress upon the significant verb. 'But I had reasons of my own which made me anxious to visit Trinidad; and I thought Barbadoes would be an excellent excuse to supply to Sir Walter for the expenses of the journey. The old gentleman jumped at it—positively jumped at it. There's nothing loosens Sir Walter's pursestrings like a devotion to business; and he declared to me on leaving, with tears in his eyes almost, that it was the first time he ever remembered to have seen me show any proper interest whatsoever in the family property.'

'And what were the reasons that made you so very anxious, then, to visit Trinidad?'

'Why, Mrs Hawthorn, how can you ask me? Wasn't I naturally desirous of seeing you and Edward once more after a year's absence?'

Marian coughed a little dry cough. 'Friendship is a very powerfully attractive magnet, isn't it, Edward?' she said with an arch smile to her husband. 'It was very good of Mr Noel to have thought of coming four thousand miles across the Atlantic just to visit you and me, dear—now, wasn't it?'

'So very good,' Edward answered, laughing, 'that I should almost be inclined myself (as a lawyer) to suspect some other underlying motive.'

'Well, she *is* a very dear little girl,' Marian went on reflectively.

'She is, certainly,' her husband echoed.

Harry laughed. 'I see you've found me out,' he answered, not altogether displeased. 'Well, yes, I may as well make a clean breast of it, Mrs Hawthorn. I've come across on purpose to ask her; and I won't go back either, till I can take her with me. I've waited for twelve months, to make quite sure I knew my own heart and wasn't mistaken about it. Every day, her image has remained there clearer and clearer than before, and I *will* win her, or else stop here for ever.'

'When a man says that and really means it,' Marian replied encouragingly, 'I believe in the end he can always win the girl he has set his heart upon.'

'But I suppose you know,' Edward interrupted, 'that her father has already made up his mind that she's to marry a cousin of hers at Pimento Valley, a planter in the island, and has announced the fact publicly to half Trinidad?'

'Not Mr Tom Dupuy?' Harry cried in amazement.

'Yes, Tom Dupuy—the very man. Then you've met him already?'

'He lunched with us to-day at Orange Grove!' Harry answered, puckering his brow a little. 'And her father actually wants her to marry that fellow! By Jove, what a desecration!'

'Then you don't like what you've seen so far of Mr Tom?' Marian asked with a smile.

Harry rose and leaned against the piazza pillar with his hands behind him. 'The man's a cad,' he answered briefly.

'If we were in Piccadilly again,' Edward Hawthorn said quietly, 'I should say that was probably a piece of pure class prejudice, Noel; but as we are in

Trinidad, and as I happen to know Mr Tom Dupuy by two or three pieces of personal adventure, I don't mind telling you in strict confidence, I cordially agree with you.'

'Ah!' Harry Noel cried with much amusement, clapping him heartily on his broad shoulder. 'So coming to Trinidad has knocked some of that radical humbug and nonsense clean out of you, has it, Teddy? I knew it would, my dear fellow; I knew you'd get rid of it!'

'On the contrary, Mr Noel,' Marian answered with quiet dignity, 'I think it has really made us a great deal more confirmed in our own opinions than we were to begin with. We have suffered a great deal ourselves, you know, since we came to Trinidad.'

Harry flushed in the face a little. 'You needn't tell me all about it, Mrs Hawthorn,' he said uneasily. 'I've heard something about the matter already from the two Dupuys, and all I can say is, I never heard before such a foolish, ridiculous, nonsensical, cock-and-bull prejudice as the one they told me about, in the whole course of my precious existence. If it hadn't been for Nora's sake—I mean for Miss Dupuy's'—and he checked himself suddenly—'upon my word, I really think I should have knocked the fellow down in his uncle's dining-room the very first moment he began to speak about it.'

'Mr Noel,' Marian said, 'I know how absurd it must seem to you, but you can't imagine how much Edward and I have suffered about it since we've been in this island.'

'I can,' Harry answered. 'I can understand it easily. I had a specimen of it myself from those fellows at lunch this morning. I kept as calm as I could outwardly; but, by Jove, Mrs Hawthorn, it made my blood boil over within me to hear the way they spoke of your husband.—Upon my honour, if it weren't for—for Miss Dupuy,' he added thoughtfully, 'I wouldn't stop now a single night to accept that man's hospitality after the way he spoke about you.'

'No, no; do stop,' Marian answered simply. 'We want you so much to marry Nora; and we want to save her from that horrid man her father has chosen for her.'

And then they began unburdening their hearts to Harry Noel with the long arrears of twelve months' continuous confidences. It was such a relief to get

a little fresh external sympathy, to be able to talk about it all to somebody just come from England, and entirely free from the taint of West Indian prejudice. They told Harry everything, without reserve; and Harry listened, growing more and more indignant every minute, to the long story of petty slights and undeserved insults. At last he could restrain his wrath no longer. 'It's preposterous,' he cried, walking up and down the piazza angrily, by way of giving vent to his suppressed emotion; 'it's abominable! it's outrageous! it's not to be borne with! The idea of these people, these hole-and-corner nobodies, these miserable, stupid, ignorant noodles, with no more education or manners than an English ploughboy—O yes, my dear fellow, I know what they are—I've seen them in Barbadoes—setting themselves up to be better than you are—there, upon my word I've really no patience with it. I shall flog some of them soundly, some day, before I've done with them; I know I shall. I can't avoid it. But what on earth can have induced you to stop here, my dear Teddy, when you might have gone back again comfortably to England, and have mixed properly in the sort of society you're naturally fitted for?'

'I did,' Marian answered firmly; 'I induced him, Mr Noel. I wouldn't let him run away from these miserable people. And besides, you know, he's been able to do such a lot of good here. All the negroes love him dearly, because he's protected them from so much injustice. He's the most popular man in the island with the black people; he's been so good to them, and so useful to them, and such a help against the planters, who are always trying their hardest to oppress them. And isn't that something worth staying for, in spite of everything?'

Harry Noel paused and hesitated. 'Tastes differ, Mrs Hawthorn,' he answered more soberly. 'For my part, I can't say I feel myself very profoundly interested in the eternal nigger question; though, if a man feels it's his duty to stop and see the thing out to the bitter end, why, of course he ought in that case to stop and see it. But what does rile me is the idea that these wretched Dupuy people should venture to talk in the way they do about such a man as your husband—confound them!'

Tea interrupted his flow of indignation.

But when Harry Noel had ridden away again towards Orange Grove on Mr Dupuy's pony, Hawthorn and his wife stood looking at one another in

dubious silence for a few minutes. Neither of them liked to utter the thought that had been uppermost in both their minds from the first moment they saw him in Trinidad.

At last Edward broke the ominous stillness. 'Harry Noel's awfully dark, isn't he, Marian?' he said uneasily.

'Very,' Marian answered in as unconcerned a voice as she could well summon up. 'And so extremely handsome, too, Edward,' she added after a moment's faint pause, as if to turn the current of the conversation.

Neither of them had ever observed in England how exceedingly olive-coloured Harry Noel's complexion really was—in England, to be as dark as a gipsy is of no importance; but now in Trinidad, girt round by all that curiously suspicious and genealogically inquiring society, they couldn't help noticing to themselves what a very dark skin and what curly hair he happened to have inherited.

'And his mother's a Barbadian lady,' Edward went on uncomfortably, pretending to play with a book and a paper-knife.

'She is,' Marian answered, hardly daring to look up at her husband's face in her natural confusion. 'He—he always seems so very fond of his mother, Edward, darling.'

Edward went on cutting the pages of his newly-arrived magazine in grim silence for a few minutes longer; then he said: 'I wish to goodness he could get engaged and married offhand to Nora Dupuy very soon, Marian, and then clear out at once and for ever from this detestable island as quickly as possible.'

'It would be better if he could, perhaps,' Marian answered, sighing deeply. 'Poor dear Nora! I wish she'd take him. She could never be happy with that horrid Dupuy man.'

They didn't dare to speak, one to the other, the doubt that was agitating them; but they both agreed in that half-unspoken fashion that it would be well if Harry pressed his suit soon, before any sudden thunderbolt had time to fall unexpectedly upon his head and mar his chance with poor little Nora.

As Harry Noel rode back to Orange Grove alone, along the level bridle-path, he chanced to drop his short riding-whip at a turn of the road by a

broad canepiece. A tall negro was hoeing vigorously among the luxuriant rows of cane close by. The young Englishman called out to him carelessly, as he would have done to a labourer at home: 'Here you, hi, sir, come and pick up my whip, will you!'

The tall negro turned and stared at him. 'Who you callin' to come an' pick up your whip, me fren'?' he answered somewhat savagely.

Noel glanced back at the man with an angry glare. 'You!' he said, pointing with an imperious gesture to the whip on the ground. 'I called you to pick it up for me. Don't you understand English?'

'You is rude gentleman for true,' the old negro responded quietly, continuing his task of hoeing in the canepiece, without any attempt to pick up the whip for the unrecognised stranger. 'If you want de whip picked up, what for you doan't speak to naygur decently? Ole-time folk has proverb, "Please am a good dog, an' him keep doan't cost nuffin." Get down yourself, sah, an' pick up your own whip for you-self if you want him.'

Harry was just on the point of dismounting and following the old negro's advice, with some remote idea of applying the whip immediately after to the back of his adviser, when a younger black man, stepping out hastily from behind a row of canes that had hitherto concealed him, took up the whip and handed it back to him with a respectful salutation. The old man looked on disdainfully while Harry took it; then, as the rider went on with a parting angry glance, he muttered sulkily: 'Who dat man dat you gib de whip to? An' what for you want to gib it him dere, Peter?'

The younger man answered apologetically: 'Dat Mr Noel, buckra from Englan'; him come to stop at Orange Grobe along ob de massa.'

'Buckra from Englan'!' Louis Delgado cried incredulously. 'Him doan't no buckra from Englan', I tellin' you, me brudder; him Trinidad brown man as sure as de gospel. You doan't see him is brown man, Peter, de minnit you look at him?'

Peter shook his head and grinned solemnly. 'No, Mistah Delgado, him doan't no brown man,' he answered, laughing. 'Him is dark for true, but still him real buckra. Him stoppin' up at house along ob de massa!'

Delgado turned to his work once more, doggedly. 'If him buckra, an' if him stoppin' up wit dem Dupuy,' he said half aloud, but so that the wondering

Peter could easily overhear it, 'when de great an' terrible day come, he will be cut off wit all de household. An' de day doan't gwine to be delayed long now, neider.' A mumbled Arabic sentence, which Peter of course could not understand, gave point and terror to this last prediction. Peter turned away, thinking to himself that Louis Delgado was a terrible obeah man and sorcerer for certain, and that whoever crossed his path, had better think twice before he offended so powerful an antagonist.

Meanwhile, Harry Noel was still riding on to Orange Grove. As he reached the garden gate, Tom Dupuy met him, out for a walk in the cool of the evening with big Slot, his great Cuban bloodhound. As Harry drew near, Slot burst away suddenly with a leap from his master, and before Harry could foresee what was going to happen, the huge brute had sprung up at him fiercely, and was attacking him with his mighty teeth and paws, as though about to drag him from his seat forcibly with his slobbering canines. Harry hit out at the beast a vicious blow from the butt-end of his riding-whip, and at the same moment Tom Dupuy, sauntering up somewhat more lazily than politeness or even common humanity perhaps demanded, caught the dog steadily by the neck and held him back by main force, still struggling vehemently and pulling at the collar. His great slobbering jaws opened hungrily towards the angry Englishman, and his eyes gleamed with the fierce light of a starving carnivore in sight and smell of his natural prey.

'Precious vicious dog you keep, Mr Dupuy,' Harry exclaimed, not over good-humouredly, for the brute had made its teeth meet through the flap of his coat lappets: 'you oughtn't to let him go at large, I fancy.'

Tom Dupuy stooped and patted his huge favourite lovingly on the head with very little hypocritical show of penitence or apology. 'He don't often go off this way,' he answered coolly. 'He's a Cuban bloodhound, Slot is; pure-blooded—the same kind we used to train in the good old days to hunt up the runaway niggers; and they often go at a black man or a brown man—that's what they're meant for. The moment they smell African blood, they're after it, like a greyhound after a hare, as quick as lightning. But I never knew Slot before go for a white man! It's very singular—ex-cessively singular. I never before knew him go for a real white man.'

'If he was my dog,' Harry Noel answered, walking his pony up to the door with a sharp lookout on the ugly mouth of the straining and quivering

bloodhound, 'he'd never have the chance again, I can tell you, to go for another. The brute's most dangerous—a most bloodthirsty creature. And indeed, I'm not sentimental myself on the matter of niggers; but I don't know that in a country where there are so many niggers knocking about casually everywhere, any man has got a right to keep a dog that darts straight at them as a greyhound darts at a hare, according to your own confession. It doesn't seem to me exactly right or proper somehow.'

Tom Dupuy glanced carelessly at the struggling brute and answered with a coarse laugh: 'I see, Mr Noel, you've been taking counsel already with your friend Hawthorn. Well, well, in my opinion, I expect there's just about a pair of you!'

(To be continued.)

TOBACCO CULTIVATION.

THE question of the cultivation of tobacco has recently been brought within the range of practical agriculture. In both Houses of Parliament the government has announced that permission will be given to grow this plant, and cure it in proper manner, as experiments, in various parts of the country, and more especially in Ireland. The Council of the Royal Agricultural Society of England, with His Royal Highness the Prince of Wales in the chair, determined to help the government in the matter, provided the government gave a grant towards the experiments. The subject thus becomes one of special moment. It is very doubtful, however, whether any experiments that can be made will give us much more information than we at present have regarding this crop. That it can be grown in this country is certain. To take up the first seed catalogue that comes to our hand—that of Messrs Carter & Co.—we find that for a long series of years past, the seed of no fewer than seven varieties of *Nicotiana* is announced as for sale. The plants are grown in many gardens, and the leaves are dried and used as fumigants against insects. In fact, so simple is the growth of the plant, that the only directions given are to ‘Sow on heat, and transplant to good, rich, loamy soil, or sow out of doors in May.’ That the plant can be grown is certain; but if grown on an agricultural scale, it will have to bear with the usual effects of climate, injurious insects, and the thousand-and-one ills which plant-life is heir to. That is, so far as the plant is concerned. The great difficulty in every country will begin with the curing, and is the cause of the tobacco crop being gradually given up.

So far as Europe is concerned, there has been a great decrease in tobacco cultivation during recent years. In the Netherlands, the acreage is at present something like half what it was ten or twelve years ago. In Belgium, the decrease in area has been considerable, but not to so great an extent. In Austro-Hungary the acreage under tobacco was in 1884 less by 8768 acres than two years previously. In Germany, the area of the crop fell from 1881 to 1883 by over 12,000 acres. Italy, with its magnificent climate, grows only 8202 acres; while in France, where the government purchase the crop, only 32,800 acres were grown last year. It is to America, however, that we

must turn for our best information as to the growth of tobacco. In the last four census years, this crop was grown to the following extent: 1850, crop of 199,752,655 pounds; 1860, crop of 434,209,461 pounds; 1870, 262,735,341 pounds; and 1880, crop of 472,661,117 pounds, grown on 638,841 acres. Here we find that although there was a great decrease in the growth of this crop after the war, it gradually picked up again, and the crop is now as large as ever. In 1883, 451,545,641 pounds were grown on an area of 638,739 acres. Its total value was £8,091,072.

The method of cultivation adopted in the United States cannot fail to be of use to the English or Irish grower. In the first place, a word should be said upon the position of tobacco in crop rotations. Travellers in South America have often noticed the desolate appearance of some portions of the country. This is due to the exhaustion of the soil by continuous tobacco-growing. A very large proportion of what was known as tobacco land has thus been reduced to a condition of poverty, and has been left to itself, and is covered with weeds. A good authority declares that this fault can be easily remedied, and that by growing tobacco as a rotation crop. After two crops of tobacco have been taken from the land, and after this a crop of corn, and then a crop of clover or vetches, after the latter have been cut or fed off, the land may be again prepared for another crop of tobacco. A word may be said here also on manures. In the best tobacco plantations, two hundred pounds of nitrate of soda and two hundred and fifty pounds of superphosphate per acre are used—the former bringing up a heavier crop, and the latter improving its quality. Besides these, large applications of farmyard manure are made. Taking Wisconsin as the State more particularly to be treated of, we find that the seed-beds are burned lightly, and a liberal allowance of manure worked in, to the depth of six inches, with a hoe or spade. This work of preparation begins in July, when the manure is applied. The bed is reworked in August, and again in September, for the purpose of keeping down any weeds or grass that may spring up; and finally, in November, it is hoed and raked and prepared to receive the seed, which is either sown in the Fall or early in the succeeding spring. When sown in the Fall, the seed is not previously sprouted. After sowing, the bed is compacted by rolling, tramping, or clapping with a board. The plants are carefully nursed by liquid manuring and by weeding. The young plants are generally large enough for transplanting by the 1st of June.

The land for the main crop—that is, into which the plants are transplanted from the seed-bed—is ploughed in the Fall after the crop of the previous year, and twice in the spring—in May, and just before the 1st of June. Coarse and rough manures are applied with the autumn ploughing, and finer well-rotted sorts in May. After the last ploughing, the land is thoroughly pulverised by harrows or drags, and marked off for the plant. The varieties of tobacco grown are either the seed-leaf or the Spanish. If the former, the plants are placed two and a half feet by three feet apart; but if the latter, three feet by a foot and a half. Thus, if the seed-leaf variety, some five thousand five hundred plants are used to the acre; and if the Spanish, nine thousand six hundred. As soon as the soil is in proper condition to work after the plants have been set out, a cultivator with five teeth is run between the rows, and this is kept up once or twice a week, until the field has been gone over five or six times. The crop is hoed twice—once after the cultivator has been run through the first time. Very little earth is put round the plant, level cultivation being preferred. In some portions of the district, a horse-hoe is used in cultivating the crop; this implement, from its peculiar construction, enables the operator to go very near each plant and stir every portion of the soil. In very small patches, the cultivation is done entirely with the hoe, which is kept up every week until the plants are so large that they cannot be worked without breaking the leaves.

The next operations are termed ‘topping’ and ‘suckering.’ In about forty-eight or fifty days after the plants are set, if the crop has been well cultivated and the weather seasonable, the flower-buds make their appearance, and are pinched out, leaving from fourteen to sixteen leaves on each plant. None of the bottom leaves are taken off, but all are left to mature, or dry up, serving as a protection against the dirt. Fields, however, are often seen in full blossom before the tobacco is topped, and this results in great damage to the crop. Tobacco is suckered twice—once in about a week after it is topped, and again just before it is cut, which is generally about two weeks after topping. ‘Suckering’ consists in the removal of young suckers, which at this time make their appearance in large numbers. As has been noted, tobacco is generally ready for harvesting in two weeks after being topped; but there is considerable variation in the time on various soils. On warm sandy loams, the plant will be as ripe in twelve days as it will be on heavy clayey soils in eighteen days. This is one of the reasons why sandy loams are preferred.

Harvesting commences early in August, and continues without intermission into September. The time preferred for cutting is from two o'clock in the afternoon until nearly sundown, because at that time tobacco is less liable to be blistered by the heat of the sun. The instrument used for cutting is a hatchet, the plants being cut off nearly on a level with the ground, and laid back on the rows to 'wilt.' After wilting, they are speared on laths. Of the large seed-leaf variety, only about six plants are put on a lath, but of the smaller Spanish (or Havana) variety, ten are not considered too many. After being speared on the laths, the latter are carefully put on a long wagon-frame, made for the purpose, and carried to the sheds, where they are arranged on the tier poles or racks, from six to ten inches apart, according to the size of the plant, but never so close as to permit them to touch each other. It requires six weeks to cure the Spanish variety perfectly, and two months to cure the seed-leaf. If the weather is dry, after the crop is out, the doors are kept closed during the day and opened at night; but extreme care must be taken not to cure too rapidly. In muggy, sultry weather, as much air as possible should be given, thorough ventilation being indispensable, to prevent 'pole-sweat.' Continuous damp weather and continuous dry weather are both to be feared. It is believed by many good growers that white veins are the result of a drought after the tobacco has been harvested, and it is said that no crop cured when there is plenty of rain is ever affected with them. Inferences of this kind, however, are too often drawn without considering a sufficient number of cases to warrant the enunciation of a general law. This is the view put forth by Mr Killebrew, in an able paper on Tobacco-culture written for the American government. He, however, further points out that it is a well-established truth, deduced from the universal experience of the cultivation of seed-leaf tobacco in every State, that a crop cannot be cured without the alternations of moist and dry atmospheres.

A few words may be said on the curing of tobacco generally. Three systems are adopted in the United States. It may be (1) air-dried; (2) dried by open-fire heat from charcoal or wood fires in the barn; or (3) by flues which convey heat from ovens and heaters built outside the barn. The last method is said to be the best, as a better control can be had over the temperature. No regular rule can be given, as the heat must be regulated according to circumstances, and must change with the weather. The main thing is to dry the tobacco gradually to secure a good colour, and to prevent mould. When the tobacco is dry, it must be kept so by gentle fires in wet or damp weather,

and it is not touched for the purpose of 'bulking' until it has become soft and pliable. Artificial sweating is believed by some to be accompanied with less risk than sweating by the natural process; and second stories of warehouses are sometimes prepared as sweating chambers by being closely ceiled or plastered. These are heated by furnaces, and the temperature maintained at from one hundred and ten to one hundred and forty degrees.

After curing, the tobacco is prepared for market. This consists of stripping the leaves from the stalks, tying them up in large bundles, and afterwards sorting them. After being sorted in 'grades,' these are tied up in 'hands' of from eighteen to twenty leaves, securely wrapped with a leaf at the butt-end, and 'bulked' in piles, with the heads out and the tails overlapping in the centre of the bulk. Here it remains until the 'fatty stems' are thoroughly cured, when it is sold to the dealers. These latter pack it in barrels and sweat the leaves still further; but into this subject we need not go, as it can have but little interest to the farmer who intends growing tobacco in this country.

So far as the cost of growing tobacco is concerned, a large and successful grower in Pennsylvania, some two years ago, published the following statement of cost and returns from a field of nine and a half acres: 215¾ days' labour of men from preparing the seed-bed up to the hanging in the barn, £43; team-work, 38½ days, with feed for 42½ days, £30; curing, stripping, and marketing, £15: total, £88. The net receipts were £174; thus showing a profit of £86. This was in a fairly good year.

These few notes show us that tobacco is a crop requiring a great expenditure of labour and care, and that even in America the profits of thirty pounds per acre, about which we have heard so much, are not always realised. The probabilities, however, are so much against our getting really fine qualities of tobacco, that it is doubtful if the necessary capital will be put into the business.

‘WHERE IGNORANCE IS BLISS.’

I WRITE these pages as a warning. I don’t suppose any one will profit by it. From the time of Cassandra downwards, nobody has ever paid attention to warnings. But that is not my affair.

A London newspaper, some years ago, gave up several columns of its valuable space to the question: ‘What shall we do with our Boys?’ I perused the correspondence with a strong personal interest, for I myself am the proprietor of a boy—several boys, in point of fact; but I refer more particularly to my eldest, aged nineteen, as to whom I felt that it was time something was settled. I have a great belief—partly derived from the before-mentioned correspondence, and partly from my own observation—in studying a boy’s natural bent, and finding him an occupation in accordance with it. Such being the case, I began to study Augustus with a view to finding out his special aptitude; but, unless a really remarkable faculty of outgrowing his trousers may be so regarded, I could not for some time discover that he had any. By dint, however, of careful observation and cross-examination of the household, I elicited that he was addicted to making extremely offensive smells in the back kitchen with chemicals, and that he had what he called a ‘collection’ of beetles and other unpleasant insects stuck on pins in a box in his bedroom. It appeared, therefore, that his proclivities were scientific, and I ultimately decided to make an analyst of him. Accordingly, after disposing of sundry painful but presumably necessary arrangements as to premium, Augustus was duly articulated to a Public Analyst. I use capital letters, because I observed that Mr Scrutin himself always did so. Why, I cannot say. Possibly, a public analyst—without capitals—would not command the same amount of public confidence. On consideration, I don’t suppose he would.

Augustus’ first demand on taking up his new occupation was a microscope. ‘And while you’re about it,’ he suggested, ‘it had better be a good one.’ At first, I was inclined to suspect that this was an artful device for the further indulgence of his entomological vices, and that the implement would be devoted to post-mortem examinations of deceased caterpillars or other

kindred abominations. He assured me, however, that such was not the case, and that the microscope was nowadays ‘the very sheet-anchor of analytical science.’ The ‘sheet-anchor’ completely took the wind out of my sails. (I feel that there is rather a confusion of metaphor here, but, not being a nautical person, I don’t feel competent to set it right.) I surrendered, humbly remarking that I supposed a five-pound note would cover it. The youthful analyst laughed me to scorn. The very least, he assured me, that a good working microscope could be got for would be ten or twelve pounds. Ultimately, I agreed to purchase one at ten guineas, and congratulated myself that at anyrate *that* was done with. On the contrary, it was only just begun. No sooner had my analyst secured his microscope, than he began to insist upon the purchase of a number of auxiliary appliances, which, it appeared, no respectable microscope would be seen without. He broke them to me by degrees. At first he only mentioned, if I remember right, an ‘achromatic condenser,’ at two guineas. Next came a ‘double nosepiece’ (why ‘double,’ I don’t know); then a polarising apparatus and a camera lucida (four pounds ten); then a micrometer and a microtome (three guineas more); then somebody’s prism, at one pound five; and somebody else’s microspectroscope, at I don’t know how much. Here, however, I put my foot down. *I* am compelled to regard the sordid consideration of price, though science doesn’t.

The microscope and its subsidiary apparatus were duly delivered; but my analyst appeared to be in no particular hurry to convey them to the laboratory where he was studying. On my making a remark to this effect, he replied: ‘Haven’t taken them to the laboratory? No; and I’m not going to. Mr Scrutin has got a precious sight better microscope than mine—cost sixty guineas without the little extra articles, and they were about thirty more. *He’s* got a microspectroscope, if you like!’

I refrained from arguing the point, and mildly remarked that in that case he might have used Mr Scrutin’s microscope, and saved me some twenty guineas. But he rejected the idea with scorn, and explained that *his* microscope was not for laboratory use, but for ‘private study.’

So far as my observation went, my analyst’s private study had hitherto been confined to a short pipe and the last number of some penny dreadful; but I did not think it wise to check his new-born ardour; I contented myself by observing that I only hoped he would ‘stick to it.’

‘No fear of that,’ he rejoined, as indignantly as a limpet might have done in answer to the same observation. ‘Why, microscopy is the most fascinating study out.—Just take a squint at *that*, now.’

I looked down the tube, but couldn’t see anything at all, and made a remark to that effect.

‘Oh, that’s because you haven’t got the focus.—*Now*, try again.’

I tried again, and saw a sort of network of red fibre.

‘I’ll bet sixpence you can’t tell me what *that* is!’ he exclaimed triumphantly.

I owned the soft impeachment.

‘That’s the maxillary gland of a rat.’

‘Dear me!’ I said.

‘Yes. Isn’t it lovely? Here’s another.—Now, just look at that.’ (A queer granular-looking object.) ‘You don’t know what that is?’

‘Give it up,’ I said.

‘That’s a section of the epidermis of the great toe.’

‘Great toe!’ I exclaimed in disgust. ‘What on earth have analysts got to do with great toes?’

‘Oh, nothing particular,’ he said airily. ‘But we like to have as much variety as possible. I should like to have a section of everything, if I could get it.—Here’s another pretty slide; that is the section of a diseased potato; and this one is a bit of a frog’s leg.’

‘Very instructive, I daresay,’ I remarked; ‘but I hope you haven’t made me spend twenty pounds merely to improve your acquaintance with frogs’ legs and diseased potatoes. Mr Scrutin surely doesn’t analyse such things as these?’

‘I can’t say we do much in frogs’ legs,’ he said; ‘but there are lots of things adulterated with potato. Flour and arrowroot, and butter, and cocoa, and—and—a heap of things. And the potato’s just as likely to be diseased as not. It *may* be, anyhow, and there you are! If you don’t know what diseased potato looks like, you’re done.’

‘A pleasant lookout,’ I replied, ‘if half-a-dozen of the commonest articles of food are habitually adulterated.’

‘Bless you, that’s nothing,’ he replied. ‘If *that* was all, there wouldn’t be much harm done. There are a jolly sight worse adulterations than that. In fact, pretty nearly everything’s adulterated, and some of ’em with rank poisons.’

‘Rank poisons! That’s manslaughter!’

‘O no; it isn’t,’ he calmly rejoined. ‘Of course, they don’t put in enough to kill you right off. And if you find something disagreeing with you, you can’t swear what it is. It *may* be the nux vomica in the beer; but it’s just as likely to be entozoa in the water, or copper in the last bottle of pickles. However, you’re all right *now*. With an analyst in the family, at anyrate you shan’t be poisoned without knowing it. I’ll let you know what you are eating and drinking.—This fellow’—and he patted the microscope affectionately—‘will tell you all about that.’

And it did. From that day forth I have never enjoyed a meal, and I never expect to do so again. I have always been particular to deal at respectable establishments, and to pay a fair price, in the hope of insuring a good article. I have, or had, a very tolerable appetite, and till that dreadful microscope came into the house, I used to get a good deal of enjoyment out of life. But now all is changed. My analyst began by undermining my faith in our baker. Now, if there was one of our tradesmen in whom, more than another, I had confidence, it was the baker, who supplied what seemed to me a good, solid, satisfying article, with no nonsense about it. But one day, shortly after the conversation I have recorded, my analyst remarked at breakfast-time: ‘We had a turn at bread yesterday at the laboratory—examined five samples; and found three of ’em adulterated. And do you know’—holding up a piece of our own bread and smelling it critically—‘I rather fancy this of ours is rather dicky.’

‘Nonsense!’ I cried. ‘It’s very good bread—capital bread!’

‘You may think so,’ he continued calmly; ‘but you’re not an analyst. I shall take a sample of this to the laboratory, and you shall have my report upon it.’

‘Take it, by all means. But if you find anything wrong about that bread, I’ll eat my hat!’

‘Better not make rash promises. I’ll take a good big sample, and you shall have my report on it to-night.’

On his return home in the evening, he began: ‘I’ve been having a go-in at your bread. It’s not pure, of course; but there isn’t very much the matter with it. There’s a little potato, and a little rice, and a little alum; and with those additions, it takes up a good deal more water than it ought, so you don’t get your proper weight.’

‘Ahem!’ I said, ‘if that’s the case, we’ll change our baker. I’m not going to pay for a mixture of potatoes and water, and call it bread. But as for alum, that’s all nonsense. If they put *that* in, we should taste it.’

‘O no; you wouldn’t. When alum is put in bread, it decomposes and forms sulphate of potash, an aperient salt. It disagrees with you, of course, but you don’t taste it. As for changing your baker, the next fellow you tried might be a jolly sight worse; *he* might put in bone-dust, or plaster of Paris, or sulphate of copper. And besides, half the adulterations are in the flour already, before it reaches the baker. Of course, that doesn’t prevent his doing a little more on his own account.’

And with that the matter dropped, so far as the bread was concerned; but my confidence was rudely shaken.

A few days later, my analyst remarked: ‘I don’t think much of this milk;’ and he forthwith appropriated a sample for analytical purposes; but, happily, was compelled to own that it wasn’t quite so bad as he expected. It had more than its proper proportion of water; but that *might* arise—he charitably suggested—from the cow being unwell. To make up the deficiency, it had been fortified with treacle and coloured with arnatto, but these my analyst appeared to regard as quite every-day falsifications.

‘It’s a rascally shame,’ I said. ‘If one can’t put faith in the milk-jug, it’s a bad lookout for the Blue Ribbon gentlemen. However, let us hope that the tea and coffee are all right.’

‘Not likely!’ he rejoined. ‘Nearly all tea is “faced,” as they call it, more or less, and the facing is itself an adulteration. As for coffee, you don’t expect to get *that* pure, do you? It’s sure to be mixed with chicory, anyhow, and very probably with roasted acorns, beans, mahogany sawdust, or old tan. Baked horse-liver occasionally; but that’s an extreme case. If by any remote chance there wasn’t anything wrong in the original coffee, you get it in the chicory; and very often there are adulterations in both; so you get ’em twice over.’

‘If that’s the case, no more ground coffee for me. We’ll grind our own, and then we are sure to be safe.’

‘You mustn’t make too cocksure of that. Some years ago, an ingenious firm took out a patent for a machine to mould chicory into the shape of coffee-berries. Smart chaps those! And of course they can put anything they like into the chicory before they work it up.’

‘That’s pleasant, certainly. Then how is one to secure pure coffee?’

‘You can’t secure it, except by sending a sample to us, or some other shop of the same sort, to have it analysed; and if it’s wrong, prosecute your grocer for adulteration. After doing that a few times, he might find it didn’t pay, and give it up.’

‘And how much would that cost?’

‘Analysis of a sample of coffee, one guinea; analysis of butter, five guineas; analysis of milk, one guinea; analysis of tea, one guinea. Those are the regular charges for private analyses.’

‘Rather expensive, it seems.—And how much would it cost to prosecute?’

‘Ah, that I can’t tell you,’ said my analyst. ‘Another fiver, or more, I daresay.—But look at the satisfaction.’

I did look at it, but ultimately decided to give my grocer the benefit of the doubt, and cherish a fond hope that he was better than his fellows. The subject dropped. But a few days later, there chanced to be apple-pudding on the table. With the dish in question my analyst had always been in the habit of consuming brown sugar, and a good deal of it. Now, however, on the sugar-basin—best Demerara—being offered to him, he put on an expression as if he had been invited to partake of black draught.

‘Raw sugar! No, thank you.’

‘Hillo, what’s wrong with the sugar? Is that adulterated too?’

‘Very probably,’ he loftily replied. ‘But that’s a small matter. The genuine article is bad enough.’

‘Bad enough!’ indignantly interposed my analyst’s mamma. ‘That’s Mr Grittles’s very best moist—threepence-three-farthings a pound!’

‘I daresay it is. If it was fourpence, it wouldn’t make any difference.—Did you ever hear of the sugar-mite, *Acarus sacchari*’——

‘No; I can’t say I ever did,’ I said, ‘and I don’t want to, either. We have had enough of this sort of thing, and I am not going to have any more agonies over every article we eat.’

I had again put my foot down. But it was too late. I had even forbidden my analyst, under penalty of forfeiture of his pocket-money for several months to come, telling us anything whatever about the food we eat or the drink we imbibe; but the mischief was done. I have lost my confidence in my fellow-man, and still more in my fellow-man’s productions. I may try in an imperfect way to protect our household. I may give the strictest orders that none but the refinedest of sugar shall be admitted into our store-cupboard; but who is to answer for the man who makes the jam and the marmalade, or the other man who makes the Madeira cakes and the three-cornered tarts? And how much is there that we have *not* heard? I have silenced my analyst’s lips, it is true; but there is also a language of the eyes, and still more a language of the nose, and when, with a scornful tip-tilt of the latter, he says, ‘No, thank you,’ to anything, my appetite is destroyed for that meal. I can’t take a pill or a black draught without my disordered imagination picturing my chemist ‘pestling a poisoned poison’ behind his counter. I can’t even eat a new-laid egg or crack a nut without wondering what it is adulterated with. This is morbid, no doubt. I am quite aware that it is morbid, but I can’t help it. I am like Governor Sancho in the island of Barataria: my choicest dishes are whisked away from me—or rendered nauseous, which is as bad—at the bidding of a grim being who calls himself Analytical Science. He may not know anything about it, or he may be lying; but meanwhile he has spoilt my appetite, and the dish may go away untasted for me.

Truly, a little knowledge is a dangerous thing. The moral of my painful story is obvious. I intend to bring up the rest of my family, if possible, to occupations involving no knowledge whatever.

THE MONTH: SCIENCE AND ARTS.

ABOUT two years ago, we recorded an interesting discovery which had been made on the coast of Norway, that of a viking war-ship, which had formed the tomb of some forgotten Danish freebooter. We have now to chronicle a somewhat similar find, which has recently been unearthed at Brigg, in Lincolnshire. While the workmen were excavating the ground for a new gas-holder, they came upon a block of oak, which ultimately proved to be an ancient British vessel of extraordinary size. It is cut out of a solid piece of wood, and measures forty-eight feet in length, fifty-two inches in width, and thirty-three inches in depth. The boat is in a wonderfully good state of preservation, owing, no doubt, to the clayey nature of the soil in which it lies, and which has effectually sealed up every cranny against the intrusion of the air. The discovery of this prehistoric relic is of such interest, that it is to be hoped some way of preserving it from the action of the weather will be found before it is too late.

Only a few years ago, an ancient wooden causeway was discovered in the same neighbourhood—a causeway made of squared balks of timber fifteen feet long and ten inches square. The ends of these logs were bored with holes for the reception of pegs, so that the whole structure could be firmly fastened to the earth. This was evidently a necessary precaution; for the causeway crosses the valley of the river Ancholme, and would be subject to removal by the action of the tidal waters. It is believed that an extensive shallow lagoon once existed in the Ancholme valley, and that this was slowly filled up with alluvium. It is to this silting up with a non-porous soil that the preservation of both the boat and the causeway is due.

The *Times of India* raises a curious point about a certain meteor of unusual brilliancy which was seen in India on a certain night in January last. Curiously enough, a meteor which was described by eye-witnesses in almost the same language which was used by the Indian observers, passed over London on the same evening. It was travelling in an easterly direction, and appeared about two hours and a half before the meteor noted in India.

The question raised by this double appearance is: Are these two meteors really one and the same? The distance between the two points of observation is between five and six thousand miles, which would give a rate of movement for the meteor of thirty-five and a half miles per minute. The question is a startling one, which we should think could be easily answered by consulting the logs of various vessels which were near the presumed track of the meteor on the night of its occurrence. Such an unusual appearance could not fail to have been recorded.

The celebrated Christy Ethnographical Collection has now been added to the British Museum, and for the first time it may be said that the country which has the best opportunities of studying prehistoric and semi-barbarous peoples in all the countries of the world, is not behind its neighbours in its collection of objects for promoting that study. Mr Henry Christy, who died in 1865, left his wonderful collection to four trustees, to deal with it as they might think fit in the best interests of science. These trustees offered the collection to the national Museum on the very wise condition, that it was not to become the property of the Museum until it should be publicly exhibited there. This proviso has prevented the collection being packed away into cellars for an indefinite time, a fate which has befallen too many treasures intrusted to the national Museum.

The delegates of the French Chambers of Commerce who accompanied M. de Lesseps during the late survey of the Panama Canal works, have now returned with hopeful tales of the ultimate success of the grand project for uniting the Atlantic and Pacific Oceans. Briefly put, the matter stands thus: let money be supplied, and the work can be brought to a glorious termination. M. de Lesseps affirms that the canal can be opened for traffic as soon as 1889; and he points to the circumstance that all contracts expire in 1888. But contractors are but mortal, and it is believed by experts that the hard Culebra rocks, which present the most formidable obstacle to the prosecution of the work, cannot be cut through in less than five years. These rocks are more than a mile in length, and in some spots they rise to a height of more than one hundred and fifty feet above the canal level.

In a recent article on 'The National Egg-supply,' a contemporary gives some interesting particulars regarding the productiveness of different kinds of fowls. The laying power of each hen is said to be on an average one hundred eggs per annum. This seems a small average. Some fowls will lay

as many as two hundred and twenty per annum, but the larger proportion yield not more than from sixty-five to one hundred and twenty per annum. Care and proper food have much to do with productiveness, as all keepers of fowls know well. A large portion of our egg-supply comes from Ireland, where the birds are not nearly so well tended as they are in England and Scotland. A score of Irish eggs selected at random from a large crate weighed a little under two pounds. The eggs from good Dorkings will weigh six ounces more than this. The eggs from Spanish fowls weigh two pounds fourteen ounces per score; while those from Leghorns weigh as much as three pounds for the same quantity. The total cost of our annual egg-supply is calculated to be nearly seven millions sterling.

Mr W. K. Brooks, of the John Hopkins University of America, has put forward a new observation regarding oyster spat, which may account for the failure of the fisheries in many parts of this country. He remarks that the young oyster as it settles upon the bottom of the sea is in some localities so covered with sediment that it is killed at a very early stage of existence. He holds that the tender oyster should find a resting-place which must be clean as well as free from destructive pests. He recommends the employment of floating frames furnished with a bottom of galvanised wire-netting for the reception of the fry. Under such conditions, it is found that oysters grow with wonderful rapidity.

Anglers know well that the voracious pike is a fish most tenacious of life, and that hours after he has lain in the fishing-creel apparently dead, he is quite capable of giving a snap with his sharp teeth. But few are aware how long a pike will live out of his proper element. A Paris fishmonger recently received a quantity of fish from Rotterdam which were packed in ice. Among these was a pike over two feet long, which, on unpacking, was seen slightly to move its gills. The fish was placed in fresh water, with the result that in a few hours it was fully alive and very active. This fish, as far as can be learnt, was actually out of the water for three days, during which time it travelled nearly three hundred miles. It is now in the Trocadero Aquarium, and seems to have fully recovered from its curious experience.

The *Sanitary Record* informs metropolitan householders that their peace is threatened with a new danger. A London resident found that each time the water was turned on to his house, a plentiful supply of coal-gas was delivered gratis at the same time and through the same pipes. The

explanation of the matter is as follows: in the particular street where this strange thing happened, the soil round the water main is completely saturated with gas from leaky pipes. When the water is turned off, there is a vacuum formed in the main, and gas is sucked in through imperfect joints, to be delivered to the unfortunate residents directly the water is again turned on. The matter can of course be easily remedied; but the serious lesson taught by the incident is that gas can find its way to water-pipes, and that sewer-gas may as easily do so as coal-gas.

The last application of rock-oil is a petroleum engine, which we saw working lately in London. In general appearance, it is like a gas engine; but it has a tank fixed above the cylinder which contains a supply of petroleum. This liquid is conveyed by a small pipe and pump to the cylinder at the rate of about four drops per stroke of the piston rod. It is ignited by a spirit-lamp after having been mingled with sufficient air to form an explosive mixture. The working cost of the engine is calculated at three-halfpence per horsepower per hour for petroleum, and one-sixth of that sum for lubricating. The engine will be valuable where gas is not to be obtained and where steam is inadmissible.

Mr William Anderson lately delivered an interesting lecture before the Royal Institution 'On New Applications of the Mechanical Properties of Cork to the Arts.' He showed that cork was unique among solid substances in being capable of cubical compression both from forces applied in opposite directions and from pressure from all sides. This is shown when cork is immersed in water and is subjected to hydraulic pressure. The phenomenon in question is due to the peculiar cellular structure of the material, which causes it to behave more like a gas when under pressure than like a solid. Mr Anderson proposes to use cork instead of air in the air-vessels of water-raising machinery, and he showed by experiment how well fitted it was for doing this duty. He also proposes to use it in connection with gun-carriages in the following way: the carriage is to be furnished with hydraulic compressors in the customary manner, but the water in the cylinders is to be driven by the recoil of the gun into a vessel filled with cork. This will represent a store of energy which will run the gun out again when loaded, by the aid of a tap which will liberate the water from the compressed cork. The lecture certainly exhibited cork in a new character, and called attention to many ways in which it can be used with advantage.

The nebula in the Pleiades, so strangely discovered by photography, although it was quite invisible to ordinary telescopic scrutiny, has now been detected by more than one observer. It is, however, as may be guessed, an extremely faint object. MM. Perrotin and Thollon, of the observatory at Nice, say that they have seen it, but admit at the same time that this was only because they knew from the Paris photograph that it existed.

The number of valuable substances which can be extracted from coal-tar is marvellous, and would surprise gas manufacturers of a generation ago, who gladly gave away the tar to any one who would take it. The last product of the black and ill-smelling fluid is a substance which has been named Saccharin, on account of its extreme sweetness, and the discovery is due to Professor Fahlberg. Saccharin is said to be two hundred and thirty times sweeter than the best cane-sugar. It has a great interest for the medical profession, for it can be used to render palatable the food of patients suffering from diabetes, and has been already adopted for this service in one of the Berlin hospitals. At present, the new sweetener costs forty shillings per pound. It has been ascertained by experiment that saccharin is innocuous; and we may feel sure that if its price can be reduced, it will become a formidable rival to sugar.

The chief of the United States Geological Survey, Major Powell, has discovered near California what he believes to be the oldest human habitations on the American continent. The mountains in the vicinity are covered with beds of lava, in which have been excavated square rooms, lined with a kind of cement made with lava. Although these rock-dwellers were of prehistoric time, their work shows traces of an advanced civilisation. Several articles of pottery have been found in these cave-dwellings, as well as a kind of cloth made of woven hair. Wrapped in such a cloth, which tumbled into dust when touched, there was found a small image resembling a man. No fewer than sixty groups of these villages in the lava have been found.

Mr Eric S. Bruce, who has been experimenting during the past year for the government with a balloon for signalling purposes, which he has invented, is about to exhibit a balloon of the same kind at the Crystal Palace, Sydenham. This aërostat will have a capacity of eighty thousand cubic feet, sufficient to give it the necessary lifting power to carry up several passengers. The balloon will be a captive one, like that exhibited at Paris in

1878, and will, like its huge forerunner, be hauled down to the earth after each ascent, by steam-power. It will ascend for the amusement of visitors during the daytime, telephonic communication being maintained between the car and the earth; while at night it will be illuminated by the electric light, so that Mr Bruce's method of signalling may be fully demonstrated.

The number of deep wells sunk in London and its neighbourhood during the past thirty years has had the effect of lowering the general water level in the chalk to the amount of about twelve inches annually. But there is still a very large quantity available—so the experts say—without sinking shafts to extraordinary depths. Much interest attaches to the subject at the present time on account of the threatened action of the London corporation to sink wells for themselves, as the strongest protest they can offer against the high charges of the Water Company supplying the city.

The title of one of Turner's best pictures, 'The *Téméraire* towed to her last Moorings,' comes to the mind as one hears that the *Great Eastern*, the largest steamship ever built, too large, indeed, to be profitably worked, has steamed round to Liverpool to serve as a show-place during the Maritime Exhibition there. After this last duty is done, this monument of Brunel's wonderful skill will take up her position as a coal-hulk.

People who rejoice in the possession of wealth and who have plenty of time on their hands, generally develop into 'collectors.' Coins, pictures, books, china, orchids, postage-stamps, &c., have their periods as the fashionable things to gather together. The last craze of this kind is devoted to engraved plates. Old copper plates are perhaps the best; and the way to preserve and exhibit them is as follows: the plate is rolled with ink and polished, just as if an impression were required of it. It is then set aside for the ink to dry, when it receives a coating of clear varnish, to protect it from the oxidising action of the air. It is now framed and hung up like an ordinary picture.

The Kyrle Societies have seldom reason to congratulate iron manufacturers on the progress of their art; but it seems as if they might heartily rejoice in a Report recently made at the instance of the North-eastern Steel Company as to the utilisation of an important by-product of the steel manufacture. The Report is on the results of experiments made to test the value of basic cinder as a manure, and is the joint work of Professor Wrightson and Dr Munro, of the College of Agriculture, Downton, Salisbury. Basic cinder, or

basic steel slag, is the broken-up and useless lining of the converters used in the Thomas-Gilchrist process for dephosphorising iron, and is a bulky by-product of the manufacture. It contains from sixteen to nineteen per cent. of phosphoric acid combined with lime and other bases; and the Report in question puts it beyond a doubt that the undissolved phosphates of the cinder have an available and remarkable value for manurial purposes. Extensive and elaborate experiments conducted at Downton and elsewhere showed decisively that this heretofore inconvenient substance is an excellent fertiliser for swedes and other turnips, as well as for grass. It seems to be positively better for this purpose than ground coprolites, and only a little less effective than superphosphate. This interesting Report is published at the *Daily Exchange* Offices, Middlesborough. Similar experiments have been attended with like success in Germany; and from *Le Temps* it would appear that enterprising agricultural chemists are already in treaty with some of the blast-furnaces of Alsace-Lorraine for the purchase of all the slag produced by them.

The history of the recovery of a portion of the mails from the Cunard steamer *Oregon*, ought to supply chemists and inventors with a good deal of food for thought. Before the vessel sank, a portion of the mail was recovered, but by far the greater portion went down with her. This was the case with the registered letters, the portion of the mail containing securities, coupons, &c., to the value of at least one hundred thousand pounds, besides drafts, letters of credit, &c., of which the value was unknown. A notice has been issued by the Liverpool postmaster which tells us that the whole of these registered letters have been recovered. The letters were thoroughly soaked, but the post-office authorities dried them as carefully as they could and sent them on to their destination. All the mail-matter that has been recovered was badly damaged by wetting, while the bags which were subjected to long-continued soaking at the bottom of the sea were very much damaged. In one case, a fifty pound note sent from Frome to Chicago was delivered only just recognisable, but still sufficient to insure its being honoured.

These facts have led an American scientific journal to urge the necessity for waterproof mailbags, waterproof paper, and waterproof ink. Waterproof mailbags alone will not be sufficient, as, in the process of handling them or raising them from a sunken vessel, they are liable to be rendered leaky.

Waterproof paper, again, would be of no service unless it was accompanied by waterproof ink. The mailbags need only be waterproof in the ordinary acceptation of the term; and if there could be certainty that they would remain so, nothing more would be needed to protect documents or anything else placed in them; but as holes are likely to be worn or torn in them, the only final resource is the production of paper and ink that will resist the prolonged action of sea-water. If such a paper and ink can be produced at a reasonable cost, they would meet with a ready market throughout the civilised world. But the paper must be lighter, more flexible, and more opaque than the waterproof parchment paper now obtainable.

The lesson which the loss of the *Oregon* seems to teach the commercial world is, that a convenient waterproof paper is required for transatlantic correspondence. Modern chemistry and mechanical invention ought to be able to meet this want.

No class of the community has received so much good advice as that to which the farmer belongs, and it would be a wonder if he did not resent some of it, and say that it is not good advice that is wanted most, but good seasons. However, when a practical lesson comes within one's reach for the better utilisation of available material, only a foolish person would neglect to learn it. In the model dairy at the Brighton Show, last summer, Professor Long gave an explanatory demonstration of the simple methods of making three kinds of soft cheese, by the employment of tinned-iron hoops, beech-boards, straw-mats, milk-vessels, draining-shelves, and a thermometer. In the Journal of the Bath and West of England Society, he has recently drawn attention to the subject again, and explains his method whereby the farmer may utilise his skim-milk by the profitable manufacture of soft cheese. It seems that anybody can learn the processes; and a few experiments will teach the practice of ripening the cheeses in an apartment having a regulated temperature proper for the development of the necessary white mould, followed by blue mould, producing the most accepted flavour.

From a gallon of 'whole' milk, costing sixpence, Professor Long made Brie cheese—the most famous of French varieties—worth, at ten days to three months old, from one shilling to one shilling and sixpence; from half a gallon of milk, half of it skim-milk, valued at twopence-halfpenny, he made Coulommiers, a round cheese worth at least eightpence; and from skim-milk only, costing about one penny, he made a square variety, of his own

invention, named Graveley cheese, partaking of the qualities of the Limburg of Germany and the Livarot of France. We understand that nearly six millions of the delicious Brie cheeses are made annually in certain districts of France for the Parisian market. An important point will be gained, however, in this country, if some of our farmers begin to convert their skim-milk into a product which will sell at three or four times the value of the milk.

Honey-wine is said to be excellent; and Dzierzon—one of the most famous German writers on scientific bee-keeping—tells us that it is often manufactured by peasants in Eastern Europe. It is made as follows: Twenty-five pounds of honey are mixed with four and a half gallons of water in a bright copper boiler, the mixture being gently boiled and constantly skimmed during half an hour. Three pounds of finely powdered chalk are then gradually added, under constant stirring. The tough scum which rises to the surface is skimmed off, and when no more rises, the liquid is poured into a wooden vessel, where it is allowed to settle. The liquid is then carefully decanted into the cleaned kettle, mixed with six pounds of finely powdered and recently burned charcoal, and raised to boiling. It is now once more poured into the wooden vessel, allowed to cool, and then filtered through felt or flannel. It should be stated that the chalk is added to neutralise free acid, while the charcoal removes the waxy taste. The filtered liquor is then transferred to the boiler, mixed with the white of twenty-five eggs, and raised to boiling, when the coagulated albumen will have clarified the liquid. After having kept the liquid at a gentle boil for one hour longer, it is allowed to cool, and is then poured into a cask, which must not be quite full, and the bung-hole covered with a piece of clean linen. In this condition it is allowed to remain until fermentation has been completed. When it is perfectly clear, the liquid is drawn off into bottles. We are told by Dzierzon that this wine, if properly prepared, resembles the best brands of Madeira, and is a truly royal beverage. It keeps for any length of time, provided the bottles are stored in a cool cellar.

A NEW THEORY OF DEW.

THE explanation of the formation of dew and hoar-frost which Dr Wells published about seventy years ago, has been almost universally accepted as satisfactory ever since. Shortly stated, Dr Wells' 'Theory of Dew' is as follows: Air always contains a certain amount of moisture in the form of invisible vapour. The hotter the air is, the more vapour will it contain. Thus, during a warm day, a good deal of moisture passes into the air; and when the temperature falls in the evening, some of it is deposited as a fine mist. But even when this mist does not appear, dew is formed. As soon as the sun is down, especially if it is a clear evening, the grass, trees, shrubs, and even the soil itself rapidly get cooled by radiating into space the heat which they contain. These cooled bodies in turn cool the warm air above them, and this causes it to deposit more or less of its moisture, which appears either as a film or in minute drops of dew. The points of the grass, small twigs, and all other good radiating surfaces are cooled the most; and accordingly we find the dewdrops most abundant on these bodies; whilst on metal or hard stone surfaces, which are poor radiators, we seldom or never find any dew. A clear, cloudless sky, which promotes radiation, is always favourable to the formation of dew; but on cloudy nights, little is formed, because the clouds return the heat radiated from the earth.

Hoar-frost is only dew deposited on bodies cooled below the freezing-point. It is formed in winter when the temperature of the air during the day is only a little over this point. At night, the grass and ground are soon cooled below thirty-two degrees Fahrenheit, and what moisture is deposited appears as minute ice-crystals or hoar-frost.

Many experiments can be cited which tend to strengthen and confirm this explanation. Thus, every one is familiar with the fact of glass bottles, mirrors, &c., being covered with moisture on being brought into a warm room. The same thing happens with a cold cabbage leaf, or with a bundle of vegetables or a bunch of flowers. On a cold night, the windows of a warm room soon get dimmed. Still more striking is a phenomenon which frequently occurs in countries where the temperature is much below the

freezing-point in winter. The houses are well heated, and if a number of people are together, as in a ballroom, the air soon becomes moisture-laden. If the ventilation is not over-good, it may happen that a door or window will be opened. With the rush of cold air from without, the merry-makers are often alarmed by being suddenly covered with hoar-frost, or sometimes even a shower of snow. This does not come from the outside, as it occurs most readily on cold, clear, starlit nights. It is the moisture of the air of the room suddenly cooled below freezing-point that appears as snow or hoar-frost. Many similar experiments may be noticed, all of which are satisfactorily accounted for on Dr Wells' theory.

Yet, within the last few months, Mr Aitken, in a communication to the Royal Society of Edinburgh, has brought forward many observations, and the results of numerous experiments, which appear to prove that Dr Wells' theory of dew is not, after all, correct.

The essential difference between the old and the new theories is as to the source of the moisture which forms the dew. Instead of being condensed from the air above by the cooled vegetation, Mr Aitken maintains that it comes from the ground. The author of the original theory admitted that some of the dew might come from below, but affirmed that it must be an exceedingly small proportion. Mr Aitken's experiments, on the contrary, seem to prove that most if not the whole comes from the ground.

It is quite clear that the grass and soil do get rapidly cooled on dewy nights; but if they are below the temperature of the air above, the ground just under the surface is much warmer. Thermometers placed on the surface of grass were often found ten to eighteen degrees lower than those placed under the surface among the stems. In such circumstances, vapour must be rising from the soil, and part of it will condense on the grass, which has been cooled by radiation. By carefully weighing small squares of turf cut from a lawn before and after the appearance of dew on them, it was always found that they *lost* weight. If the dew had condensed out of the surrounding air, the turf would have *gained* in weight by the amount of dew deposited. It was thus clear that vapour was rising from the ground, only part of which was condensed on the grass, the remainder passing into the air.

Another experiment, pointing to the same conclusion, was made by inverting thin trays over the grass. On dewy nights these trays were always

found wet on the under surface; and the grass below them was always much wetter than that freely exposed outside. The moisture rising from the ground was evidently trapped and condensed, instead of being allowed to pass freely into the atmosphere.

The explanation of the absence of dew on the surface of stones, roads, and other hard surfaces, on the old theory was, that these, being poor radiators, did not get much cooled. But closer observation shows that dew does form on stones and clods and gravel, only it is chiefly on the under surfaces. Thus, slates laid over both hard and gravelly roads are always found dripping wet on their under surfaces on dewy nights; while their upper surfaces and the surrounding roads are dry. During frost, too, clods and stones on the surface of the soil are almost always found to be covered with hoar-frost, showing that the moisture is trapped as it rises from the soil.

But perhaps the most interesting observations and experiments were those made to determine the origin of the 'dewdrops' on grass and vegetables. In the first place, it is found that these drops do not appear on all plants. Some are wet, while others growing alongside are dry, though there could be no great difference in their radiating power. Then the leaves do not get wet all over, but only at the edges and on the tips. A closer observation reveals the fact that these so-called 'dewdrops' are formed at the end of the minute veins of the leaves and grass, and are not now recognised as dew at all, but moisture exuded from the interior of the plants themselves. Moreover, these drops always appear before the true dew in the evening, and very often are seen when no true dew is formed. They even appear when the vegetables are placed under conditions where condensation of the surrounding water-vapour is impossible, and must, therefore, be due to the vital activity of the plants.

Another observation may be mentioned which clearly shows that moisture rising from below may become condensed on the cooled surfaces of loose material. If the weather is at all cold, the beard and moustaches get covered with moisture; and in very cold climates, the eyebrows, hair, and whiskers get covered with a coating of hoar-frost. The moisture which forms this certainly comes from the body, which is always at a much higher temperature than the surrounding air.

All these observations and experiments have led to the conclusion that moisture is constantly being given off from the earth; and that, except on the rare occasions when a warm moisture-laden wind blows gently over a previously cooled surface, it only returns to the surface of the ground after being condensed into rain, sleet, snow, or hail. Dew is only a portion of the outward current trapped on the exposed and cooled surfaces of the grass and other bodies.

COMRIE EARTHQUAKES.

REGARDING earth-tremors or earthquakes, which, curiously enough, seem to be mainly confined in Scotland to Comrie, in Perthshire, a correspondent writing from Comrie kindly favours us with the following notes as to the erection which is there devoted to the registering of earthquakes. Our correspondent says:

I recently visited the building with a view of giving you a few notes as to its history and construction. I may state that about fourteen years ago, the British Association applied to Mr Drummond of Drumearn for leave to erect a house on his property, which he at once granted free of charge, and assisted to defray the cost of erection.

The reason why the British Association selected a site here and erected this earthquake-house at Comrie, was on account of the long-continued periodical shocks that had been felt in Upper Strathearn, particularly from the year 1780 to 1848. About the former date, they had been rather severely felt over the whole district, and damage to some extent done to buildings. On a sheet of water near to Lawers House, the ice was shattered to pieces. Some of the inhabitants at that time kept a record of their occurrence; and we believe the late Sir David Dundas, of Dunira, had a seismometer placed on his estate in Glenlednoch, to the north of Comrie; but there seems to be no evidence to show that it had ever indicated any shock. Coming down to the year 1839, the inhabitants of the village of Comrie were greatly alarmed, about eleven o'clock on the night of the 23d of October, by one of the most violent tremors that had been experienced there; and the good people rushed out of their houses and assembled in the old Secession Church for prayer, which was conducted by the Rev. R. T. Walker, the minister of that church. Many others fled to the hills. But no serious damage was done to property, save some rents in the chimneys. From 1839 to 1847, tremors continued to be more frequent, causing considerable alarm by the movements of furniture and crockery.

The work of erecting the building proposed by the British Association was carried out under the care of the late Dr James Bryce of Glasgow, who

resided here for many seasons, and was well acquainted with the locality and its geological formation. The site chosen is a rising ground near Drumearn House, and is built on rock that is supposed to extend a considerable distance westward. The building is stone, and slated, and is about seven feet square inside. The floor is laid with Arbroath pavement, on solid rock, and is overlaid with fine sand, on which are placed two boards, at right angles to each other. These boards are six feet long by nine inches broad, and on each are placed, standing, nine round wooden pins, varying from the fourth of an inch to an inch and a half in diameter, but all of one height (eight inches).

The building is in excellent condition, and the pins or markers are in their places, awaiting the action of an earthquake to record the desired information as to the severity and direction of this now seemingly extinct agency of force in Upper Strathearn. The size of pin or cylinder thrown down, and the direction in which it falls, indicate the strength of shock as well as its direction. Any one who feels interested and may wish to visit the building will readily get access by applying to Mr Drummond.

Many theories have been propounded as to the cause of the earthquakes which have visited this district. The late Mr Patrick M'Farlane of Comrie, who took a great interest in them, erected a seismometer in the steeple of the parish church of Comrie, which was visited by many of the members of the British Association and others; but so far as we are aware, it never registered any markings. It was a very simple apparatus. The pendulum was of considerable length, and all but rested on a table overlaid with magnesia, which, being light, offered no resistance to the oscillation of the pendulum. A few slight shocks occurred between 1847 and 1877, but these attracted little notice.

I may remark that no earthquake had, till recently, been felt here for some years, consequently, there had been no registering. But on Sunday morning 18th April last, at one o'clock, and again on Thursday the 22d of the same month, about half-past five A.M., a slight earthquake occurred. I visited the earthquake-house on both occasions; but there were no markings, none of the pins having fallen.

WHICH?

If thou art false as thou art fair,
And false the fairest fair may be,
Again the wondrous power to snare,
Again the siren's self we see.
There's danger in those dimpling smiles,
It glances from that witching e'e,
And he who would escape thy wiles,
Must quickly from the tempter flee.

For better far, as sages tell,
From fickle fair to bid adieu,
Than fall beneath the magic spell
Of charms the heart may ever rue.
Beware, if false, of beauty bright,
Beware that luring beacon's ray,
For, oh! the love that trusts its light,
May drift a wreck ere dawn of day.

But if thou'rt true as thou art fair,
Art leal in heart, though seeming gay,
Wouldst ever constant prove, and ne'er
With faithful heart all faithless play,
Then thou'rt a gem worth more than gold,
More precious than the ruby rare,
More to be prized than wealth untold,
True heart enshrined in form so fair.

JOHN NAPIER.

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